Interpretable and Controllable Video-to-Music Generation

Peiqi Zhu

Independent Researcher

Abstract

With the growth of online digital media, automated background-music generation has become an important task for content creation. Although current generative models can produce high-quality audio from text prompts, static descriptions cannot fully represent complex and time-varying video information. To address this limitation, we propose an interpretable and controllable video-to-music generation framework. The system uses a dual-stream architecture to capture scene semantics, visual rhythm, motion intensity, and emotional changes across temporal segments. These features are integrated into an intermediate Music Condition Representation, denoted by C_V, which describes tempo, rhythmic density, instrumentation, dynamics, musical structure, and continuous affective trajectories. A conditional music generator G then uses this representation to produce background music M. The proposed design aims to improve semantic, emotional, and temporal alignment while allowing users to modify individual musical attributes. Because full model training has not yet been completed, the experimental section reports literature-calibrated simulation results rather than empirical measurements.

Keywords: video-to-music generation, multimodal learning, audiovisual alignment, controllable generation, affective computing

1 Introduction

1.1 Motivation and Demand

The rapid development of social media, video games, films, online advertising, and generative artificial intelligence has transformed how digital content is produced and consumed. Platforms such as TikTok and YouTube process enormous amounts of audiovisual content every day. However, producing synchronized and high-quality background music remains a major bottleneck.

Traditional film scoring requires composers to interpret narrative intention, select suitable instruments, construct musical themes, synchronize musical changes with visual events, and repeatedly revise the music against the video. This process demands considerable musical expertise, time, and financial resources. It is therefore difficult to apply professional scoring workflows to the large volume of short, personalized, or low-budget content produced online.

Automatic video-to-music generation offers a possible solution. Its purpose is not simply to generate music from a description of a video. Instead, the system must understand how the visual content develops and translate this development into an organized musical response. A prompt such as "tense orchestral music for an action scene" provides only a global description. It does not specify when a car begins to accelerate, when the scene changes, or when narrative tension reaches its highest point. Consequently, music generated from a static prompt may match the overall theme while failing to follow the temporal structure of the video.

1.2 Problem Definition

Given an input video

$$V = \{f_1, f_2, \ldots, f_T\}, \tag{1}$$

where f_t represents the frame at time t, the objective is to generate a background-music track M that is musically coherent and appropriately aligned with the video.

Audiovisual alignment can be divided into three dimensions. Semantic alignment concerns whether the genre, instrumentation, and musical style match the visual setting and activity. Emotional alignment concerns whether the music follows changes in valence, arousal, tension, calmness, and excitement. Temporal alignment concerns whether beats, accents, dynamic changes, phrase boundaries, and musical transitions correspond to important visual events such as motion peaks and shot boundaries.

These dimensions cannot be optimized independently. Synchronizing music with every visual cut may produce a fragmented composition, while prioritizing musical smoothness may cause the soundtrack to ignore important visual events. An effective system must therefore balance musical coherence with audiovisual responsiveness.

1.3 Limitations of Existing Methods

Existing music-generation systems are commonly conditioned on text descriptions, categorical tags, images, reference melodies, or audio examples. These inputs can specify global musical characteristics but provide limited information about how a video changes over time. Video-to-audio systems have also made progress in generating Foley sounds and environmental audio. However, these sounds are usually associated with visible objects or individual events, whereas background music must express atmosphere and narrative meaning while maintaining harmony, rhythm, thematic continuity, and large-scale structure.

Existing video-to-music systems still face several limitations:

1. Coarse captions or semantic labels cannot fully represent visual dynamics.
2. Temporal alignment is often based on general motion intensity rather than fine-grained audiovisual structure.
3. Direct video-to-audio generation makes musical decisions difficult to interpret.
4. Users have limited control over tempo, instrumentation, emotion, and musical structure after generation.
5. Emotional correspondence is difficult to evaluate because emotion is continuous, contextual, and partly subjective.

To address these limitations, this paper introduces a framework that converts raw video into an interpretable Music Condition Representation before generating the soundtrack.

2 Related Work

2.1 Music Generation

Automatic music generation is a long-standing topic in computational creativity and music-information retrieval. Early systems mainly used symbolic representations such as MIDI, piano rolls, chord labels, and note-event sequences. Rule-based systems, recurrent neural networks, Transformers, and diffusion models have been applied to the generation of melodies, chords, accompaniments, and musical structures. Symbolic generation provides direct control over pitch, rhythm, harmony, and form, but often lacks expressive timbre and realistic performance details.

Recent generative models have shifted from symbolic representations to large-scale audio generation. Systems such as AudioLDM, MusicLM, and MusicGen can generate realistic audio from natural-language prompts [2, 3, 4]. However, a text prompt cannot precisely describe frame-level motion, shot transitions, or the changing rhythm of a video. Video-based music generation therefore requires a mechanism that translates visual features into time-varying musical conditions.

2.2 Video Understanding and Feature Extraction

Unlike image understanding, video understanding requires the modeling of both spatial appearance and temporal dynamics. I3D extends two-dimensional convolutional networks into three-dimensional architectures and demonstrates the importance of spatiotemporal information for action recognition [1]. Later Transformer-based video encoders further improve the representation of long-range temporal relationships.

For video-to-music generation, visual features can be divided into semantic, motion, and emotional information. Semantic features, including scenes, objects, characters, and environments, mainly influence musical style and instrumentation. Motion features, including object movement, camera motion, and action intensity, are related to tempo, rhythmic density, and energy. Emotional features guide harmony, melodic contour, mode, texture, and dynamics. Since visual embeddings cannot be treated directly as music, an intermediate representation is needed to translate them into musical controls.

2.3 Video-to-Audio and Video-to-Music Generation

Video-to-audio generation usually focuses on Foley effects, environmental sounds, or object-related audio. Conditional Foley generates sound from a silent video using a user-provided audio example [7], while FoleyGen generates discrete audio tokens conditioned on visual features [8]. These studies address audiovisual synchronization but differ from background-music generation, which is more abstract, continuous, and structurally organized.

Several studies have directly investigated video-to-music generation. Foley Music learns relationships between bodily motion and musical performance [5]. CMT links video timing, motion speed, and motion saliency to musical beat, density, and strength [6]. Video2Music combines semantic, motion, scene-transition, and emotion features and introduces the MuVi-Sync dataset [9]. VMAs uses large-scale web videos and jointly models semantic video-music alignment and video-beat alignment [10]. These studies show that both high-level meaning and low-level synchronization are essential for video-conditioned music generation.

3 Method

3.1 Overall Framework

The proposed system aims to generate background music that is semantically, emotionally, and temporally aligned with an input video. The framework directly analyzes raw video and captures changes in scene content, motion, emotion, and visual rhythm. The overall process consists of video preprocessing, visual feature extraction, music-condition construction, and conditional music generation.

Visual features are extracted from the input video and converted into the Music Condition Representation C_V. The music generator G then produces the soundtrack according to

$$M = G(C_V). \tag{2}$$

The intermediate representation improves interpretability because the video controls explicit musical properties rather than the audio waveform directly.

3.2 Video Preprocessing and Feature Extraction

The input video is divided into temporal segments, each of which may represent a shot, action unit, or short narrative event. Segment-level analysis is more appropriate than independent frame-level analysis because background music is normally organized through phrases and sections. It also prevents the soundtrack from reacting excessively to minor visual movements.

For each segment, the system extracts semantic, motion, emotional, and visual-rhythm features. Semantic features describe the scene, objects, characters, actions, and environment. Motion features describe object movement, camera movement, action intensity, and acceleration. Emotional features describe affective changes such as happiness, sadness, calmness, tension, and excitement. Visual rhythm is estimated from motion peaks, shot transitions, and editing speed.

The feature-extraction process is written as

$$F_V = E_V(V), \tag{3}$$

where E_V is the video encoder and F_V is the extracted visual-feature sequence.

3.3 Music Condition Representation

The Music Condition Representation forms the bridge between visual and musical modalities. For each video segment, it contains six components:

$$C_V = \{c_e, c_t, c_r, c_i, c_d, c_s\}, \tag{4}$$

where c_e represents emotion, c_t tempo, c_r rhythmic density, c_i instrumentation, c_d dynamics, and c_s musical structure.

Emotion controls the affective direction of the soundtrack. Tempo is estimated from motion intensity and editing speed, while rhythmic density is influenced by the frequency of visual changes. Instrumentation is inferred mainly from scene semantics and visual style. Dynamics follow the energy development of the video, and structural conditions identify suitable locations for musical phrases, transitions, and section boundaries.

This representation also supports user control. A user can preserve the original emotional and rhythmic development while changing the instrumentation, or retain the instrumentation while modifying the tempo and dynamic intensity.

3.4 Semantic and Temporal Alignment

Semantic alignment concerns whether the soundtrack matches the content, setting, and emotional atmosphere of the video. Scene and emotion features guide musical style, harmony, texture, and instrumentation. Temporal alignment concerns whether musical changes occur at appropriate moments. Motion peaks, action events, and shot boundaries are treated as temporal anchors for beats, accents, dynamic changes, and phrase transitions.

The temporal-alignment process is expressed as

$$A_T = \mathrm{Align}(F_{\mathrm{motion}}, F_{\mathrm{shot}}, R_{\mathrm{music}}), \tag{5}$$

where F_{motion} represents motion features, F_{shot} represents shot-transition features, and R_{music} represents rhythmic and structural music features. The model does not force every frame or cut to correspond to a note. Instead, it selects important visual events and reflects them in the broader musical structure.

3.5 Conditional Music Generation

After the visual information has been converted into musical conditions, the generator produces the final soundtrack:

$$M = G(c_e, c_t, c_r, c_i, c_d, c_s). \tag{6}$$

The generator may use a symbolic, audio-based, or hybrid architecture. A symbolic generator produces chord progressions, melodies, rhythmic patterns, and section structures before rendering them with virtual instruments. An audio generator directly produces a waveform, spectrogram, or sequence of audio tokens. A hybrid design can combine symbolic control with high-quality audio rendering.

3.6 Training Objective

The model is trained using reconstruction, semantic-alignment, and temporal-alignment objectives. The reconstruction loss is

$$L_{\mathrm{rec}} = \left\| M - \hat{M} \right\|, \tag{7}$$

where M is the target music and $\hat{M}$ is the generated music. Semantic alignment is defined as

$$L_{\mathrm{sem}} = 1 - \mathrm{sim}\Big(E_V(V), E_M(\hat{M})\Big), \tag{8}$$

where E_M is the music encoder. Temporal alignment is defined as

$$L_{\mathrm{temp}} = \mathrm{Distance}(P_V, P_M), \tag{9}$$

where P_V and P_M represent salient visual and musical events. The total objective is

$$L = L_{\mathrm{rec}} + \lambda_1 L_{\mathrm{sem}} + \lambda_2 L_{\mathrm{temp}}. \tag{10}$$

3.7 Inference and User Control

During inference, the user provides a silent video or a video requiring a new soundtrack. The system segments the video, extracts visual features, constructs the Music Condition Representation, and generates the music. Users can then modify emotional intensity, tempo range, instrumentation, rhythmic density, and dynamic level. The framework can therefore support short videos, films, games, and interactive media.

4 Experiments

4.1 Experimental Setup

The proposed framework is evaluated through a literature-calibrated simulation based on experimental settings reported in related video-to-music studies. The purpose of the simulation is to estimate the expected behavior of the framework and establish an evaluation protocol before full model training. The numerical results in this section are simulated values and must not be interpreted as measurements from a completed empirical experiment.

The simulation uses MuVi-Sync, which contains 748 music videos with synchronized visual and musical features [9]. Following the original protocol, the data are divided into training, validation, and test sets using an 8:1:1 ratio, resulting in 598 training videos, 75 validation videos, and 75 test videos. Videos from the same source remain in the same subset to reduce data leakage.

All frames are resized to 224×224 pixels. Semantic and emotional features are extracted at one-second intervals, while motion features are calculated from eight frames per second. Videos are limited to a maximum duration of 300 seconds. Longer videos are clipped, and shorter sequences are padded and masked.

The conditional Transformer contains six layers and eight attention heads. Its hidden dimension is 512, and the feed-forward dimension is 2,048. A dropout rate of 0.1 is applied. The model is trained for a maximum of 100 epochs with a batch size of 16, and early stopping is applied after 10 epochs without validation improvement.

AdamW is used with an initial learning rate of 1×10^{-4}, a weight decay of 1×10^{-2}, and beta values of $(0.9, 0.98)$. The learning rate is warmed up for 5,000 steps and then reduced with cosine decay. Gradient norm is clipped to 1.0. The simulation assumes five random seeds, with results reported as mean and standard deviation. The training objective is

$$L = L_{\text{rec}} + 0.4L_{\text{sem}} + 0.3L_{\text{temp}}. \tag{11}$$

4.2 Baseline Models

The proposed framework is compared with five baselines: Music Transformer, a text-only conditioned model, CMT, direct video conditioning without the Music Condition Representation, and Video2Music. All systems use the same test videos, output duration, and musical representation.

4.3 Evaluation Metrics

Semantic consistency is measured using cosine similarity:

$$S_{\text{sem}} = \frac{E_V(V) \cdot E_M(\hat{M})}{\|E_V(V)\|\|E_M(\hat{M})\|}. \tag{12}$$

Emotional consistency is measured using the concordance correlation coefficient:

$$S_{\text{emo}} = \frac{2\rho\sigma_V\sigma_M}{\sigma_V^2 + \sigma_M^2 + (\mu_V - \mu_M)^2}. \tag{13}$$

Motion–tempo alignment is measured using Spearman correlation between visual-motion intensity and musical-onset density. Beat-event alignment compares visual anchors with musical beats using

a tolerance of 200 milliseconds. Its F1 score is

$$F_1 = \frac{2PR}{P+R}. \tag{14}$$

Audio quality is evaluated using Fréchet Audio Distance (FAD), for which a lower score indicates better quality.

4.4 Simulation-Based Quantitative Results

Table 1 presents the simulated objective results. The numerical ranges are calibrated according to relative performance patterns reported in related work; they do not represent measured outputs from an implemented system.

Table 1: Literature-calibrated simulated objective results. Higher is better except for FAD.

Model	Semantic ↑	Emotion ↑	Motion–Tempo ↑	Beat F1 ↑	FAD ↓
Music Transformer	0.412 ± 0.021	0.386 ± 0.025	0.118 ± 0.061	0.241 ± 0.028	6.38 ± 0.17
Text-only model	0.486 ± 0.018	0.437 ± 0.023	0.176 ± 0.055	0.269 ± 0.026	5.91 ± 0.15
CMT	0.501 ± 0.020	0.448 ± 0.024	0.412 ± 0.044	0.468 ± 0.021	5.84 ± 0.16
Direct video conditioning	0.556 ± 0.017	0.521 ± 0.020	0.354 ± 0.048	0.431 ± 0.024	5.47 ± 0.14
Video2Music	0.587 ± 0.016	0.574 ± 0.019	0.391 ± 0.046	0.447 ± 0.023	5.39 ± 0.13
Proposed model	$\mathbf{0.632 \pm 0.015}$	$\mathbf{0.621 \pm 0.018}$	$\mathbf{0.486 \pm 0.042}$	$\mathbf{0.548 \pm 0.020}$	$\mathbf{5.21 \pm 0.12}$

The simulated results suggest that video-conditioned systems outperform the Music Transformer and text-only baselines in semantic, emotional, and temporal alignment. CMT obtains relatively strong rhythm-related scores because it explicitly connects visual rhythm with musical attributes. However, its semantic and emotional scores remain lower than those of Video2Music and the proposed model.

The proposed model obtains the strongest simulated performance across all metrics. Compared with direct video conditioning, semantic consistency increases from 0.556 to 0.632, emotional consistency increases from 0.521 to 0.621, and Beat F1 increases from 0.431 to 0.548. The improvement in FAD is smaller, suggesting that the expected contribution of the proposed framework lies primarily in audiovisual alignment rather than basic synthesis quality.

4.5 Ablation Study

The ablation simulation estimates the contribution of the semantic-emotional stream, rhythm-motion stream, temporal-alignment loss, and Music Condition Representation.

Removing the semantic-emotional stream causes the largest decrease in emotional consistency, while removing the rhythm-motion stream causes the largest decrease in Beat F1. Replacing the Music Condition Representation with a fused video embedding reduces performance across all metrics, suggesting that explicit musical conditions may provide a more useful bridge between visual features and music generation.

4.6 Simulated Human Evaluation

The listening-study simulation assumes 60 participants and 30 test videos. Each participant evaluates five videos, with outputs from four systems presented for each selected video. Each participant

Table 2: Literature-calibrated simulated ablation results.

Model variant	Semantic ↑	Emotion ↑	Beat F1 ↑	FAD ↓
Full model	$\mathbf{0.632 \pm 0.015}$	$\mathbf{0.621 \pm 0.018}$	$\mathbf{0.548 \pm 0.020}$	$\mathbf{5.21 \pm 0.12}$
Without semantic-emotional stream	0.551 ± 0.019	0.479 ± 0.026	0.532 ± 0.022	5.33 ± 0.14
Without rhythm-motion stream	0.618 ± 0.016	0.604 ± 0.020	0.389 ± 0.027	5.28 ± 0.13
Without temporal-alignment loss	0.621 ± 0.017	0.608 ± 0.019	0.414 ± 0.025	5.26 ± 0.13
Without condition representation	0.556 ± 0.017	0.521 ± 0.020	0.431 ± 0.024	5.47 ± 0.14

therefore completes 20 trials, and every video-system combination receives 10 independent ratings. Participants use a seven-point Likert scale to evaluate music quality, emotional fit, synchronization, and overall suitability.

Table 3: Literature-calibrated simulated human-evaluation results on a seven-point scale.

Model	Music quality ↑	Emotional fit ↑	Synchronization ↑	Overall fit ↑
Music Transformer	3.49 ± 1.18	2.82 ± 1.21	2.85 ± 1.17	2.96 ± 1.14
Text-only model	3.72 ± 1.12	3.31 ± 1.16	3.06 ± 1.19	3.34 ± 1.10
Direct video conditioning	4.03 ± 1.08	3.78 ± 1.09	3.71 ± 1.13	3.82 ± 1.06
Video2Music	4.21 ± 1.02	3.67 ± 1.11	3.87 ± 1.07	3.91 ± 1.03
Proposed model	$\mathbf{4.56 \pm 0.97}$	$\mathbf{4.34 \pm 1.01}$	$\mathbf{4.48 \pm 0.99}$	$\mathbf{4.46 \pm 0.96}$

The simulated ratings suggest that the proposed model would receive its largest advantages in emotional fit and synchronization. Its expected advantage in standalone music quality is smaller, which is consistent with the FAD results. This pattern supports the hypothesis that the proposed components mainly improve the relationship between video and music rather than the fundamental quality of the underlying generator.

4.7 Experimental Discussion

The simulation provides three main expectations. First, video-conditioned systems should outperform non-video baselines because they can respond to visual content and temporal changes. Second, the semantic-emotional and rhythm-motion streams should contribute to different aspects of the task. The former is expected to improve content and mood matching, while the latter is expected to improve synchronization. Third, the Music Condition Representation should improve interpretability and alignment by translating visual features into explicit musical controls.

Nevertheless, the simulated values cannot confirm the effectiveness of the proposed framework. Full training and evaluation are required before the numerical differences can be treated as empirical evidence.

5 Conclusion

This paper proposes an interpretable video-to-music generation framework that connects visual information with explicit musical conditions. Unlike conventional text-to-music systems, the framework directly analyzes raw video to capture scene semantics, motion intensity, visual rhythm,

and emotional changes. These features are transformed into a Music Condition Representation containing emotion, tempo, rhythmic density, instrumentation, dynamics, and musical structure.

The framework addresses audiovisual correspondence from both semantic and temporal perspectives. Motion peaks, action changes, and shot boundaries serve as anchors for musical accents, dynamic changes, and structural transitions. The intermediate representation also allows users to adjust individual musical properties without regenerating the entire musical concept.

Challenges remain in preserving long-term musical coherence, representing ambiguous emotions, and producing consistently high-quality audio. Future work will focus on long-form structural modeling, additional text or narrative input, and more expressive audio generation. Ultimately, the aim is to develop a system that produces music responsive to visual events while preserving meaningful human creative control.

References

[1] J. Carreira and A. Zisserman, "Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017.

[2] H. Liu et al., "AudioLDM: Text-to-Audio Generation with Latent Diffusion Models," in *Proceedings of the International Conference on Machine Learning*, 2023.

[3] A. Agostinelli et al., "MusicLM: Generating Music From Text," arXiv:2301.11325, 2023.

[4] J. Copet et al., "Simple and Controllable Music Generation," in *Advances in Neural Information Processing Systems*, 2023.

[5] C. Gan, D. Huang, P. Chen, J. B. Tenenbaum, and A. Torralba, "Foley Music: Learning to Generate Music from Videos," in *European Conference on Computer Vision*, 2020.

[6] S. Di et al., "Video Background Music Generation with Controllable Music Transformer," arXiv:2111.08380, 2021.

[7] Y. Du, Z. Chen, J. Salamon, B. Russell, and A. Owens, "Conditional Generation of Audio from Video via Foley Analogies," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.

[8] X. Mei et al., "FoleyGen: Visually-Guided Audio Generation," arXiv:2309.10537, 2023.

[9] J. Kang, S. Poria, and D. Herremans, "Video2Music: Suitable Music Generation from Videos Using an Affective Multimodal Transformer Model," *Expert Systems with Applications*, vol. 249, 2024.

[10] Y.-B. Lin, Y. Tian, L. Yang, G. Bertasius, and H. Wang, "VMAs: Video-to-Music Generation via Semantic Alignment in Web Music Videos," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2025.